

IU CLOUD PROGRAMMING

Secure and Globally Distributed Static Website on AWS Using Terraform

Final Product Documentation

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A private S3 origin delivered globally through CloudFront and provisioned with Terraform.

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1. Introduction, Problem, Objective, and Scope

1.1 Introduction

This project demonstrates how a single static HTML page can be delivered as a secure, highly available cloud workload without operating virtual servers. The implementation uses a private Amazon S3 bucket as the origin, Amazon CloudFront as the global delivery layer, Origin Access Control (OAC) for signed origin requests, and Terraform for the complete infrastructure lifecycle.

1.2 Problem and objective

The problem is not webpage complexity but dependable delivery. A public audience needs HTTPS, global low latency, managed scaling, and high availability, while the origin must not be directly accessible. The objective is therefore to deploy one self-contained index.html object privately in S3 and expose it only through CloudFront. The same Terraform configuration must support creation, update, verification, reproduction, and secure destruction.

1.3 Scope

- One private S3 bucket and one index.html object.
- S3 ownership controls, Public Access Block, versioning, and AES256 encryption.
- One CloudFront OAC, one distribution, and a source-restricted bucket policy.
- Terraform configuration, documentation, diagrams, tests, and genuine screenshots.

EC2, ECS, Docker, Lambda, API Gateway, databases, load balancers, Route 53, custom domains, ACM, WAF, CI/CD, React, Node.js, and backend code are outside the scope because the workload contains no server-side behavior or client-side routes.

2. Functional and Cloud Requirements

The requirements translate the assignment objective into verifiable cloud controls. Each control is connected to an implementation decision and genuine evidence rather than being treated as an unsupported design claim.

Requirement	Implementation	Verification
Publish one page	S3 object key index.html with UTF-8 HTML content type	S3 object metadata and live page
HTTPS delivery	CloudFront redirect-to-https and default TLS certificate	HTTP redirect and HTTPS response tests
Private origin	All Public Access Block values true; OAC-only policy	CLI values true and direct S3 HTTP 403
Global latency	CloudFront edge delivery and managed caching	CloudFront headers and cache-hit evidence
Managed scale	S3 and CloudFront; no fixed compute capacity	No EC2, ECS, load balancer, or autoscaling resources
Reproducibility	Terraform files and retained provider lock file	Validate passes; final plan reports no changes
Secure removal	force_destroy only for this temporary assessment project	Documented terraform destroy procedure

2.1 Acceptance criteria

Completion requires the CloudFront URL to open over HTTPS, HTTP to redirect, CloudFront to report Deployed, direct S3 access to be denied, every Public Access Block value to remain enabled, Terraform validation to pass, cache behavior to be demonstrated, and the final plan to show no unexpected changes.

3. Architecture and Request Flow

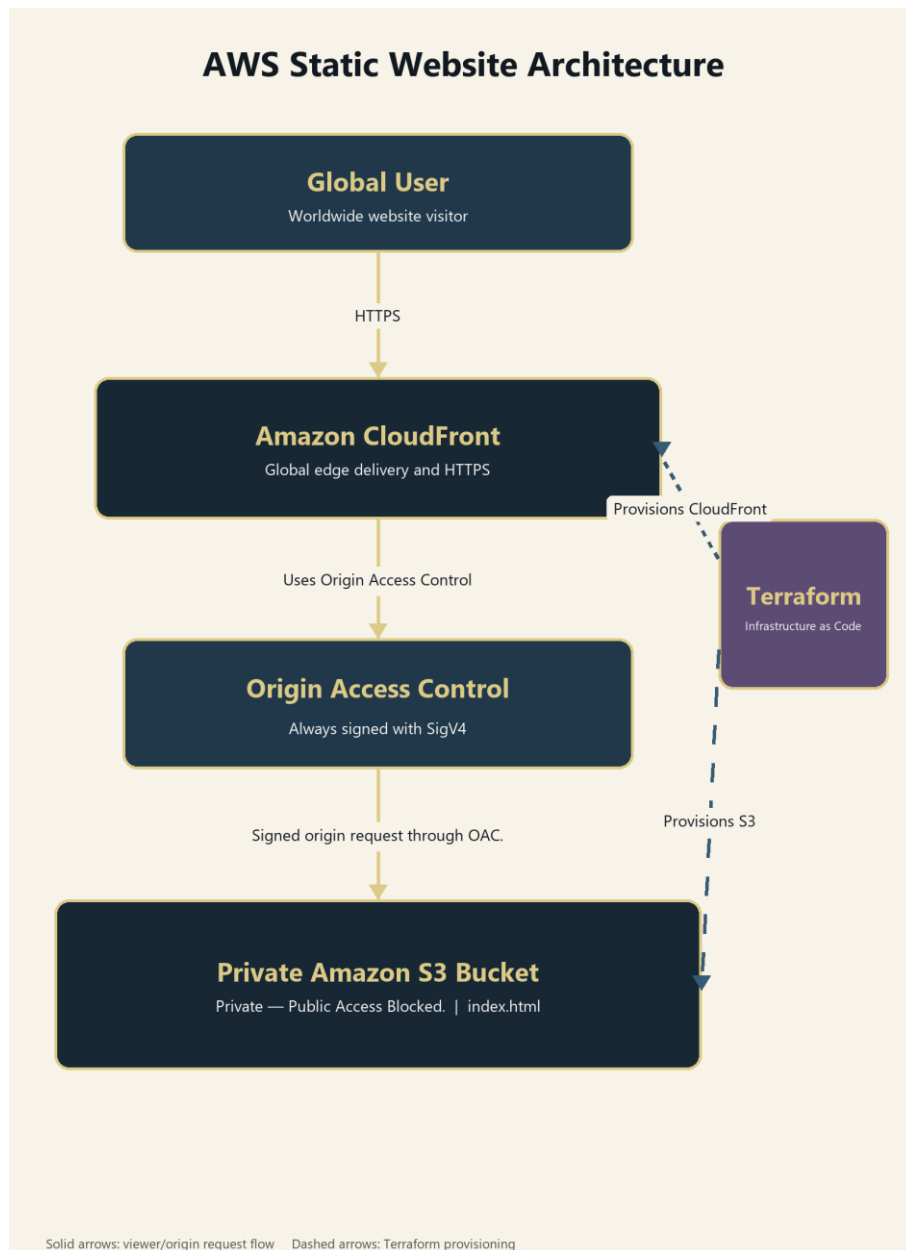


Figure 1. Global user request flow and Terraform provisioning flow.

3.1 Viewer and origin flow

1. The viewer requests the CloudFront domain. HTTP is redirected to HTTPS.
2. CloudFront checks its edge cache and returns a cached copy when available.
3. On an origin request, OAC signs the request with AWS Signature Version 4.
4. The private S3 bucket returns index.html only when the distribution-specific policy condition is satisfied.

Solid arrows in Figure 1 represent viewer and origin requests. Dashed arrows represent Terraform provisioning of the CloudFront and S3 resources.

4. AWS Service Selection and Alternatives

4.1 Amazon S3

S3 is appropriate because the origin is one static object. It provides managed, redundant object storage without an operating system, runtime, or server fleet. Versioning supports recovery, while encryption and ownership controls strengthen the storage baseline [2].

4.2 Amazon CloudFront

CloudFront provides the public HTTPS endpoint, worldwide edge delivery, compression, and managed caching [1]. PriceClass_All prioritizes the global-delivery requirement over restricting requests to lower-cost regions.

4.3 Origin Access Control

OAC signs every origin request with SigV4 and supports a bucket policy restricted by the distribution source ARN. This prevents users from bypassing CloudFront through the S3 regional URL [3].

4.4 Terraform

Terraform represents the infrastructure as reviewable configuration. The same files support planning, application, repeatable updates, and destruction, while .terraform.lock.hcl records provider selections [6].

4.5 Rejected alternatives

EC2 and a load balancer would introduce operating systems, patching, compute targets, health checks, and additional cost. Containers, Lambda, API Gateway, databases, DNS, custom certificates, and backend code are similarly unnecessary because the page has no server-side processing, dynamic data, authentication, forms, APIs, or client-side routes.

5. Terraform Design and File Structure

5.1 Source structure

```
iu-cloud-programming/  
|-- site/index.html  
|-- diagrams/architecture-diagram.png  
|-- documentation/  
|-- evidence/  
|-- environment/tool-versions.txt  
|-- main.tf  
|-- variables.tf  
|-- outputs.tf  
|-- .terraform.lock.hcl  
|-- .gitignore  
+-- README.md
```

5.2 Configuration logic

variables.tf defines eu-central-1 and validates the lowercase project name. main.tf requires Terraform 1.6.0 or later, configures the AWS provider 6.x constraint and default tags, and reads the current account ID. The project name and account ID form the globally unique bucket name. Individual resources configure S3, upload index.html, create OAC and CloudFront, and attach the restricted policy. outputs.tf exposes only non-secret deployment identifiers and URLs.

5.3 Change detection and lifecycle

The S3 object uses filemd5() so a visible HTML change creates an in-place object update. Terraform plan allows review before apply. After deployment, a second plan confirms idempotency. force_destroy is explicitly limited to this temporary student project and is not appropriate for important production data.

Stable code snapshot: [portfolio-part-2-first-draft](#)

6. Security Design and Implementation

6.1 Storage boundary

- BucketOwnerEnforced disables ACL-based ownership behavior.
- block_public_acls, block_public_policy, ignore_public_acls, and restrict_public_buckets are all true [4].
- SSE-S3 with AES256 encrypts index.html at rest.
- Versioning supports recovery from unintended replacement.

6.2 Distribution boundary

The bucket policy permits only s3:GetObject to cloudfront.amazonaws.com. The AWS:SourceArn condition must equal the ARN of the created distribution. The origin uses the regional bucket domain, not the public S3 website endpoint. OAC signs every origin request, and viewers are redirected from HTTP to HTTPS.

6.3 Secret and repository protection

AWS credentials are not hard-coded, output, committed, or documented. Authentication uses the local AWS CLI context. .gitignore excludes Terraform runtime directories, state, plans, variable files, .env files, PEM files, and private keys, while retaining .terraform.lock.hcl. The assignment PDF and knowledge-base DOCX are also excluded. Credential-pattern scans are performed before snapshots and packaging.

6.4 Temporary-project warning

The bucket uses force_destroy = true so terraform destroy can remove versioned assessment objects. This convenience must not be copied to a production bucket containing important data.

7. Deployment and Installation Procedure

7.1 Prerequisites and authentication

Install AWS CLI v2, Terraform CLI, Git, an editor, and a modern browser. Configure an authorized AWS identity outside the repository, then verify it with `aws sts get-caller-identity`. Never paste access keys into Terraform files.

7.2 Deployment workflow

1. Change to the `iu-cloud-programming` directory.
2. Run `terraform init` and retain `.terraform.lock.hcl`.
3. Run `terraform fmt -recursive` and `terraform fmt -check -recursive`.
4. Run `terraform validate` and correct every error.
5. Run `terraform plan -out=tfplan` and review the proposed resources.
6. Run `terraform apply tfplan` and wait for CloudFront deployment.
7. Run `terraform output -raw website_url` and open the HTTPS URL.

7.3 Controlled update

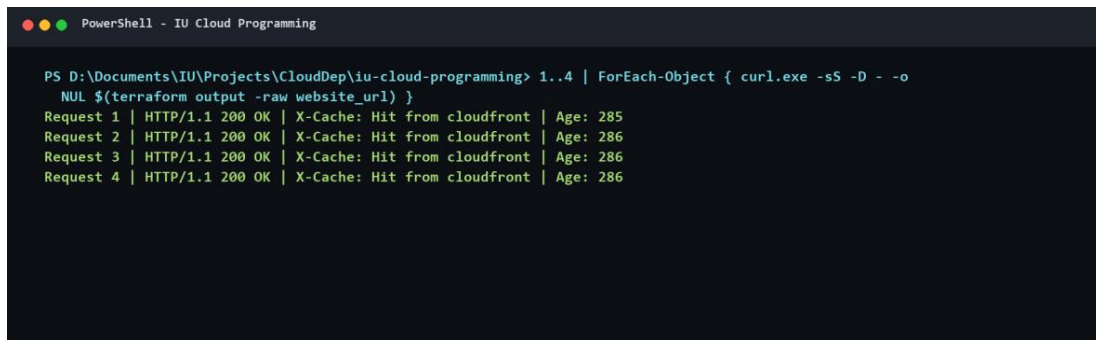
Edit `site/index.html`, create and apply a reviewed plan, then invalidate CloudFront with `aws cloudfront create-invalidation --distribution-id $(terraform output -raw cloudfront_distribution_id) --paths "/"`. Wait for completion, refresh the site, and run `terraform plan` again.

The complete command-by-command procedure, testing commands, destruction steps, troubleshooting guidance, and security warnings appear in Appendix A.

8. Testing and Evidence

Testing covers positive delivery behavior, negative origin access, configuration controls, caching, and Terraform idempotency. `evidence/test-results.md` records the command, expected result, actual result, PASS or FAIL status, screenshot, and execution date for each test.

Requirement	Implementation	Verification
HTTPS	HTTP 200 and CloudFront-related headers	PASS
Redirect	HTTP 301 to equivalent HTTPS URL	PASS
Direct S3	Unauthenticated origin request returns 403	PASS
Public access	Four Public Access Block values are true	PASS
CloudFront	Distribution status is Deployed	PASS
Caching	Repeated request shows Hit from cloudfront	PASS
Terraform	Final plan reports no changes	PASS
Redeployment	One object update and completed invalidation	PASS



```
PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> 1..4 | ForEach-Object { curl.exe -sS -D - -o NUL $(terraform output -raw website_url) }
Request 1 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 285
Request 2 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 286
Request 3 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 286
Request 4 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 286
```

Figure 2. Repeated requests demonstrate a CloudFront cache hit.

Appendix C contains all sixteen genuine screenshots, including tool versions, validation, planning, application, the live website, AWS configuration, response tests, cache behavior, idempotency, and repeatable redeployment.

9. Tutor Feedback, Corrections, and Reflection

9.1 Tutor feedback status

No tutor feedback was available at the time this document was produced. Consequently, no tutor comment, requested correction, or approval has been invented. Any later feedback should be added verbatim to the feedback matrix, linked to a concrete change, and supported by updated evidence.

9.2 Verified corrections

- Removed nested Git repository metadata after embedded-repository errors.
- Excluded the assignment PDF and knowledge-base DOCX from the repository.
- Ran Terraform commands from `iu-cloud-programming` so evidence paths resolved correctly.
- Standardized screenshots in `EvidenceScreenshots` and recorded only genuinely observed PASS results.
- Added a visible deployment-status change, applied only the object update, completed a CloudFront invalidation, and confirmed a final no-change plan.

9.3 Reflection

The most important lesson is that successful provisioning is only one part of cloud engineering. A credible result also needs negative security tests, evidence traceability, reproducible source control, and a clean destruction path. OAC and a distribution-scoped bucket policy provide stronger evidence of origin privacy than relying on a public bucket with an obscure URL. Terraform improves repeatability, but the plan must still be reviewed before every apply. The final no-change plan demonstrates that the declared configuration and deployed environment converged.

10. Results, Limitations, Future Improvements, and Conclusion

10.1 Results

The website is available through the CloudFront HTTPS endpoint. HTTP redirects to HTTPS, CloudFront reports Deployed, repeated requests demonstrate caching, direct S3 access is denied, and all Public Access Block controls are enabled. Terraform validation passes, repeatable updates work, and the final plan reports no changes.

10.2 Limitations

The implementation intentionally uses the default CloudFront domain and certificate. It has no custom domain, Route 53 record, ACM certificate, WAF, CI/CD pipeline, access-log analysis, backend application, or dynamic data. PriceClass_All favors global reach and may cost more than a restricted price class. Cache invalidation is manual.

10.3 Future improvements

- Add a custom domain and ACM certificate when domain ownership is available.
- Introduce automated validation and deployment only when the assessment scope permits CI/CD.
- Enable logging, monitoring, and budget alarms for longer-lived operation.
- Evaluate WAF only if a realistic threat model and cost justification exist.

10.4 Conclusion

The solution uses the smallest suitable architecture for one static page. S3 provides private managed storage, CloudFront supplies global HTTPS delivery and caching, OAC restricts origin access, and Terraform controls the lifecycle. The project meets its functional, availability, scaling, security, and reproducibility objectives without unnecessary compute infrastructure.

Live website: <https://d216eikm4r4k38.cloudfront.net>

Appendix A. Installation Manual

A.1 Prerequisites

- AWS account and an authorized IAM identity.
- AWS CLI version 2, Terraform 1.6.0 or later, Git, editor, and browser.
- Project files including main.tf, variables.tf, outputs.tf, .terraform.lock.hcl, and site/index.html.

A.2 AWS authentication

```
aws configure --profile iu-cloud
$env:AWS_PROFILE = "iu-cloud"
aws sts get-caller-identity
```

Use an approved profile, IAM Identity Center session, or environment-based credential mechanism. Confirm the expected account before Terraform operations. Do not save credentials in the repository.

A.3 Initialize, format, validate, plan, and apply

```
cd "D:\Documents\IU\Projects\CloudDep\iu-cloud-programming"
terraform init
terraform fmt -recursive
terraform fmt -check -recursive
terraform validate
terraform plan -out=tfplan
terraform show -no-color tfplan > evidence\terraform-plan.txt
terraform apply tfplan
```

Review the plan before applying. Expected changes are limited to S3, CloudFront, OAC, the bucket policy, and the HTML object. Do not apply a plan containing compute instances, databases, IAM users, access keys, Route 53, or load balancers.

Appendix A. Installation Manual (continued)

A.4 Retrieve and test the URL

```
terraform output -raw website_url
curl.exe -I $(terraform output -raw website_url)
curl.exe -I http://$(terraform output -raw cloudfront_domain_name)
aws s3api get-public-access-block --bucket $(terraform output -raw s3_bucket_name)
curl.exe -I https://$(terraform output -raw s3_origin_domain)/index.html
aws cloudfront get-distribution --id $(terraform output -raw cloudfront_distribution_id) --query
"Distribution.Status"
```

Expected results are HTTPS 200 through CloudFront, HTTP redirect to HTTPS, four true Public Access Block values, HTTP 403 from direct S3, and Deployed distribution status.

A.5 Update the page and invalidate CloudFront

```
# Edit site/index.html
terraform plan -out=tfplan
terraform apply tfplan
aws cloudfront create-invalidation --distribution-id $(terraform output -raw cloud-
front_distribution_id) --paths "/*"
terraform plan
```

Wait until the invalidation status is Completed, refresh the website, verify the visible change, and confirm the final plan contains no unexpected differences.

A.6 Destroy the temporary environment

```
terraform destroy
```

Review the destruction plan carefully. Confirm the CloudFront distribution, OAC, bucket policy, object, and bucket are removed. `force_destroy` is acceptable only for this temporary assessment environment.

Appendix A. Troubleshooting and Security Warnings

A.7 Troubleshooting

- Path not found when redirecting plan output: change to `iu-cloud-programming` and confirm `evidence/` exists.
- No visible output from terraform show redirection: successful redirection writes to the file; open `evidence/terraform-plan.txt`.
- CloudFront still serves old content: create an invalidation, wait for Completed, and refresh without relying on a stale browser cache.
- AccessDenied during Terraform: confirm the AWS profile, account identity, region, and required permissions.
- Bucket name conflict: retain the account-ID naming logic and verify the authenticated account.
- State lock or interrupted operation: inspect the running process and state backend before taking any recovery action; never delete state blindly.

A.8 Security warnings

- Never commit AWS access keys, secret keys, session tokens, Terraform state, plan files, `.env` files, PEM files, or private keys.
- Do not make the S3 bucket public to solve an origin-access problem. Diagnose OAC, bucket policy, and SourceArn instead.
- Do not share screenshots that contain credentials or session tokens.
- Do not run terraform destroy against the wrong workspace, account, or production environment.
- Do not use `force_destroy` for important production data.
- Re-run the credential-pattern scan before every snapshot and final package.

Appendix B. Complete Terraform Listings

The listings below reproduce the complete submitted Terraform source files. The stable, tutor-access-controlled source snapshot is also referenced from Content Page 5.

B.1 variables.tf

```
variable "aws_region" {
  description = "AWS Region containing the private S3 origin bucket."
  type        = string
  default     = "eu-central-1"
}

variable "project_name" {
  description = "Lowercase project name used in AWS resource names."
  type        = string
  default     = "iu-cloud-programming-spa"

  validation {
    condition     = can(regex("^[a-z0-9-]+$", var.project_name))
    error_message = "project_name may contain only lowercase letters, numbers, and hyphens."
  }
}
```


Appendix B. Complete Terraform Listings (continued)

B.2 main.tf

```
terraform {
  required_version = ">= 1.6.0"

  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.0"
    }
  }
}

provider "aws" {
  region = var.aws_region

  default_tags {
    tags = {
      Project      = "IU Cloud Programming"
      Environment  = "Student"
      ManagedBy    = "Terraform"
    }
  }
}

data "aws_caller_identity" "current" {}

data "aws_cloudfront_cache_policy" "caching_optimized" {
  name = "Managed-CachingOptimized"
}

locals {
  bucket_name = "${var.project_name}-${data.aws_caller_identity.current.account_id}"
  origin_id    = "private-s3-origin"
}

resource "aws_s3_bucket" "site" {
  bucket = local.bucket_name

  # Convenient for removing this temporary student project after assessment.
  # Do not use force_destroy for important production data.
  force_destroy = true
}

resource "aws_s3_bucket_public_access_block" "site" {
  bucket = aws_s3_bucket.site.id

  block_public_acls       = true
  block_public_policy     = true
  ignore_public_acls     = true
  restrict_public_buckets = true
}

resource "aws_s3_bucket_ownership_controls" "site" {
  bucket = aws_s3_bucket.site.id

  rule {
    object_ownership = "BucketOwnerEnforced"
  }
}

resource "aws_s3_bucket_server_side_encryption_configuration" "site" {
  bucket = aws_s3_bucket.site.id

  rule {
    apply_server_side_encryption_by_default {
      sse_algorithm = "AES256"
    }
  }
}

resource "aws_s3_bucket_versioning" "site" {
  bucket = aws_s3_bucket.site.id

  versioning_configuration {
    status = "Enabled"
  }
}

resource "aws_s3_object" "index" {
  bucket = aws_s3_bucket.site.id
  key    = "index.html"
  source = "${path.module}/site/index.html"
  etag   = filemd5("${path.module}/site/index.html")

  content_type = "text/html; charset=utf-8"
  cache_control = "public, max-age=300"
}
```

```

depends_on = [
  aws_s3_bucket_ownership_controls.site,
  aws_s3_bucket_server_side_encryption_configuration.site,
  aws_s3_bucket_versioning.site,
]
}

resource "aws_cloudfront_origin_access_control" "site" {
  name = "${var.project_name}-oac"
  description = "OAC for the private static website origin"
  origin_access_control_origin_type = "s3"
  signing_behavior = "always"
  signing_protocol = "sigv4"
}

resource "aws_cloudfront_distribution" "site" {
  enabled = true
  is_ipv6_enabled = true
  default_root_object = "index.html"
  price_class = "PriceClass_All"
  comment = "Secure and globally distributed static website"

  origin {
    domain_name = aws_s3_bucket.site.bucket_regional_domain_name
    origin_id = local.origin_id
    origin_access_control_id = aws_cloudfront_origin_access_control.site.id
  }

  default_cache_behavior {
    target_origin_id = local.origin_id
    viewer_protocol_policy = "redirect-to-https"
    allowed_methods = ["GET", "HEAD", "OPTIONS"]
    cached_methods = ["GET", "HEAD", "OPTIONS"]
    compress = true
    cache_policy_id = data.aws_cloudfront_cache_policy.caching_optimized.id
  }

  restrictions {
    geo_restriction {
      restriction_type = "none"
    }
  }

  viewer_certificate {
    cloudfront_default_certificate = true
  }
}

data "aws_iam_policy_document" "allow_cloudfront" {
  statement {
    sid = "AllowCloudFrontReadOnly"
    effect = "Allow"
    actions = [
      "s3:GetObject",
    ]
    resources = [
      "${aws_s3_bucket.site.arn}/*",
    ]

    principals {
      type = "Service"
      identifiers = [
        "cloudfront.amazonaws.com",
      ]
    }

    condition {
      test = "StringEquals"
      variable = "AWS:SourceArn"
      values = [
        aws_cloudfront_distribution.site.arn,
      ]
    }
  }
}

resource "aws_s3_bucket_policy" "allow_cloudfront" {
  bucket = aws_s3_bucket.site.id
  policy = data.aws_iam_policy_document.allow_cloudfront.json

  depends_on = [aws_s3_bucket_public_access_block.site]
}

```

Appendix B. Complete Terraform Listings (continued)

B.3 outputs.tf

```
output "website_url" {
  description = "Public HTTPS URL of the CloudFront website."
  value       = "https://${aws_cloudfront_distribution.site.domain_name}"
}

output "cloudfront_domain_name" {
  description = "CloudFront distribution domain name."
  value       = aws_cloudfront_distribution.site.domain_name
}

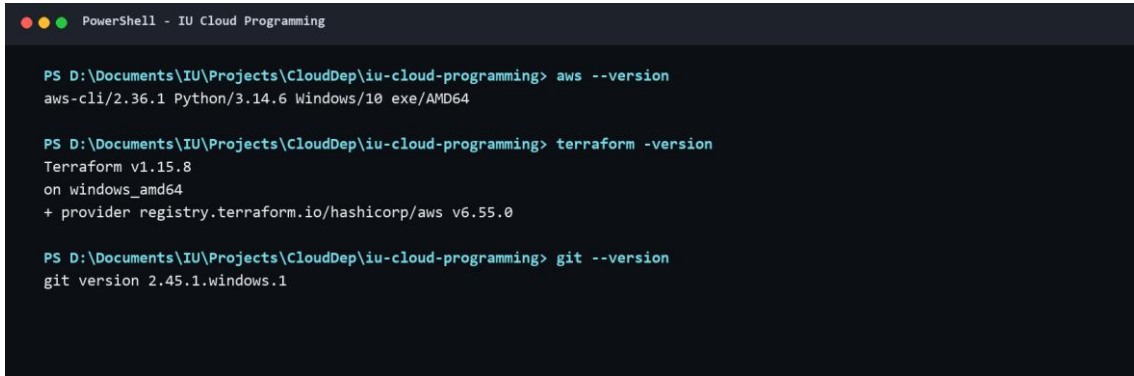
output "cloudfront_distribution_id" {
  description = "CloudFront distribution ID."
  value       = aws_cloudfront_distribution.site.id
}

output "s3_bucket_name" {
  description = "Name of the private S3 origin bucket."
  value       = aws_s3_bucket.site.bucket
}

output "s3_origin_domain" {
  description = "Private S3 regional origin domain used for access-denied testing."
  value       = aws_s3_bucket.site.bucket_regional_domain_name
}
```

Appendix C. Complete Evidence Gallery

Figures 3-18 reproduce every genuine screenshot captured for the project. Screenshots contain command and console evidence only; credential-like values were excluded.



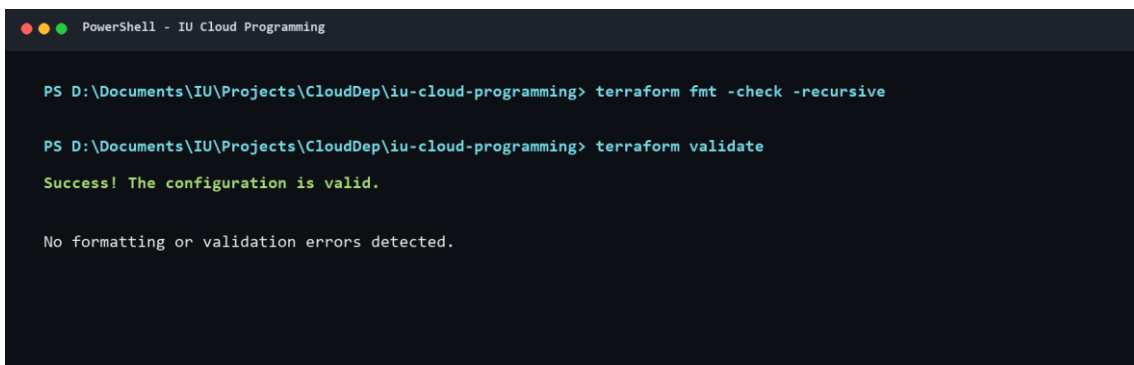
```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> aws --version
aws-cli/2.36.1 Python/3.14.6 Windows/10 exe/AMD64

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> terraform -version
Terraform v1.15.8
on windows_amd64
+ provider registry.terraform.io/hashicorp/aws v6.55.0

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> git --version
git version 2.45.1.windows.1
```

Figure 3. Installed AWS CLI, Terraform, and Git versions.



```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> terraform fmt -check -recursive

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> terraform validate
Success! The configuration is valid.

No formatting or validation errors detected.
```

Figure 4. Terraform formatting and validation completed successfully.

Appendix C. Complete Evidence Gallery (continued)

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> terraform plan -out=tfplan
Terraform will perform the following actions:

+ aws_cloudfront_distribution.site will be created
+ aws_cloudfront_origin_access_control.site will be created
+ aws_s3_bucket.site will be created
+ aws_s3_bucket_ownership_controls.site will be created
+ aws_s3_bucket_policy.allow_cloudfront will be created
+ aws_s3_bucket_public_access_block.site will be created
+ aws_s3_bucket_server_side_encryption_configuration.site will be created
+ aws_s3_bucket_versioning.site will be created
+ aws_s3_object.index will be created

Plan: 9 to add, 0 to change, 0 to destroy.

Saved the plan to: tfplan
```

Figure 5. Terraform plan created the expected resources.

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> terraform apply tfplan
aws_cloudfront_distribution.site: Creation complete after 2m34s [id=EZW53YPRFUFEA]
data.aws_iam_policy_document.allow_cloudfront: Read complete after 0s [id=377510756]
aws_s3_bucket_policy.allow_cloudfront: Creation complete after 0s
[id=iu-cloud-programming-spa-943865900195]

Apply complete! Resources: 9 added, 0 changed, 0 destroyed.

Outputs:

cloudfront_distribution_id = "EZW53YPRFUFEA"
cloudfront_domain_name = "d216e1km4r4k38.cloudfront.net"
s3_bucket_name = "iu-cloud-programming-spa-943865900195"
s3_origin_domain = "iu-cloud-programming-spa-943865900195.s3.eu-central-1.amazonaws.com"
website_url = "https://d216e1km4r4k38.cloudfront.net"
```

Figure 6. Initial Terraform apply completed successfully.

Appendix C. Complete Evidence Gallery (continued)

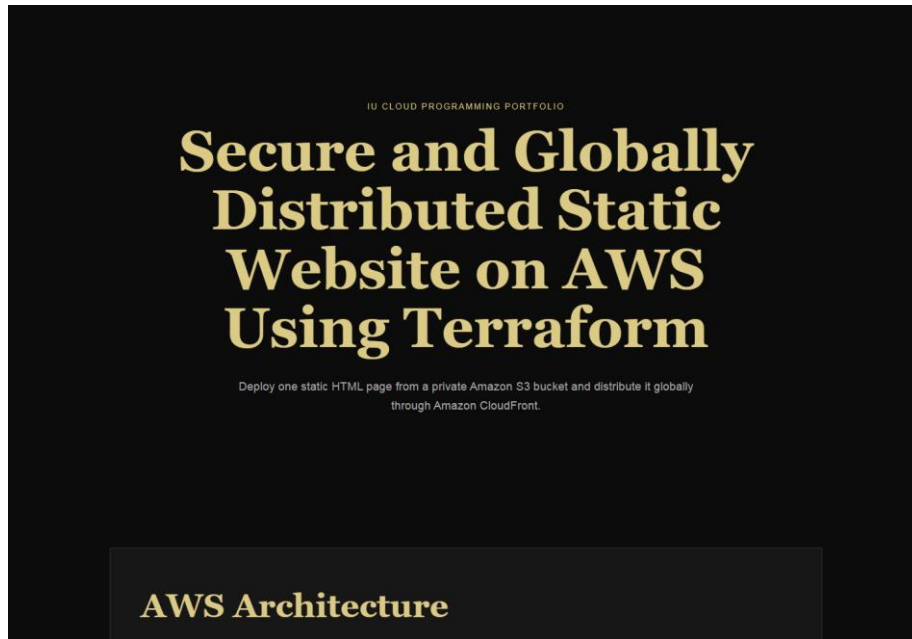


Figure 7. Live website opened through the CloudFront HTTPS endpoint.

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> aws cloudfront get-distribution --id
EZWS3YPRFUFEA --profile iu-cloud --query
"Distribution.{Id:Id,Status:Status,DomainName:DomainName,Enabled:DistributionConfig.Enabled}" --output
json
{
  "Id": "EZWS3YPRFUFEA",
  "Status": "Deployed",
  "DomainName": "d216eikm4r4k38.cloudfront.net",
  "Enabled": true
}
```

Figure 8. CloudFront distribution details show Deployed status.

Appendix C. Complete Evidence Gallery (continued)

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> aws s3api head-object --bucket
iu-cloud-programming-spa-943865900195 --key index.html --profile iu-cloud --query "{Key:'index.html',C
ontentLength:ContentLength,ContentType:ContentType,CacheControl:CacheControl,ServerSideEncryption:Serv
erSideEncryption}" --output json
{
  "Key": "index.html",
  "ContentLength": 5874,
  "ContentType": "text/html; charset=utf-8",
  "CacheControl": "public, max-age=300",
  "ServerSideEncryption": "AES256"
}
```

Figure 9. S3 index.html metadata and encryption details.

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> aws s3api get-public-access-block --bucket
iu-cloud-programming-spa-943865900195 --profile iu-cloud --output json
{
  "PublicAccessBlockConfiguration": {
    "BlockPublicAcls": true,
    "IgnorePublicAcls": true,
    "BlockPublicPolicy": true,
    "RestrictPublicBuckets": true
  }
}
```

Figure 10. S3 Public Access Block configuration in AWS evidence.

Appendix C. Complete Evidence Gallery (continued)

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> curl.exe -I $(terraform output -raw
website_url)
HTTP/1.1 200 OK
Content-Type: text/html; charset=utf-8
Content-Length: 5874
Connection: keep-alive
Date: Fri, 17 Jul 2026 20:04:17 GMT
Last-Modified: Fri, 17 Jul 2026 19:38:17 GMT
ETag: "773e7b145d42bafef13bb1eba8f7199f1"
x-amz-server-side-encryption: AES256
Cache-Control: public, max-age=300
x-amz-version-id: eZ4_4IuTS4HNKAKlbeFcHTZ2hYKn_AGa
Accept-Ranges: bytes
Server: AmazonS3
X-Cache: Hit from cloudfront
Via: 1.1 1e0407892b0e0ef93d9e2ad761e692f0.cloudfront.net (CloudFront)
X-Amz-Cf-Pop: JED50-P5
X-Amz-Cf-Id: cVvgVIz69kdLUkWN6TfmQuDFVA0oB1wmL1FgASekAcIi3E0A-IA9Uw==
Age: 151
```

Figure 11. HTTPS response includes CloudFront-related headers.

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> curl.exe -I http://$(terraform output -raw
cloudfront_domain_name)
HTTP/1.1 301 Moved Permanently
Server: CloudFront
Date: Fri, 17 Jul 2026 20:06:47 GMT
Content-Type: text/html
Content-Length: 167
Connection: keep-alive
Location: https://d216eikm4r4k38.cloudfront.net/
X-Cache: Redirect from cloudfront
Via: 1.1 8a03721ee58c2ffb24d5ec19609c4f24.cloudfront.net (CloudFront)
X-Amz-Cf-Pop: JED50-P5
X-Amz-Cf-Id: 24Vjkep-t8_MFNILb_3giqqMrLK8nd_62PQsUhEwOclH6mcD4cFRQ==
```

Figure 12. HTTP request redirects to the HTTPS URL.

Appendix C. Complete Evidence Gallery (continued)

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> aws s3api get-public-access-block --bucket
$(terraform output -raw s3_bucket_name)
{
  "PublicAccessBlockConfiguration": {
    "BlockPublicAcls": true,
    "IgnorePublicAcls": true,
    "BlockPublicPolicy": true,
    "RestrictPublicBuckets": true
  }
}
```

Figure 13. CLI verification shows all four Public Access Block values true.

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> curl.exe -I https://$(terraform output -raw
s3_origin_domain)/index.html
HTTP/1.1 403 Forbidden
x-amz-request-id: 06QKWASBV2E9XNDW
x-amz-id-2: 8eGwda05hMrdUQMy1b4gYOEh0Pf53RkyiCMerVUoj0hQf1ZSARipC+XchQCYE94o0K9qS8veYyAs9k7Va45w3VtqNo3N
jeM41fXImXbTdIY=
Content-Type: application/xml
Transfer-Encoding: chunked
Date: Fri, 17 Jul 2026 20:06:49 GMT
Server: AmazonS3
```

Figure 14. Direct unauthenticated S3 request is denied with HTTP 403.

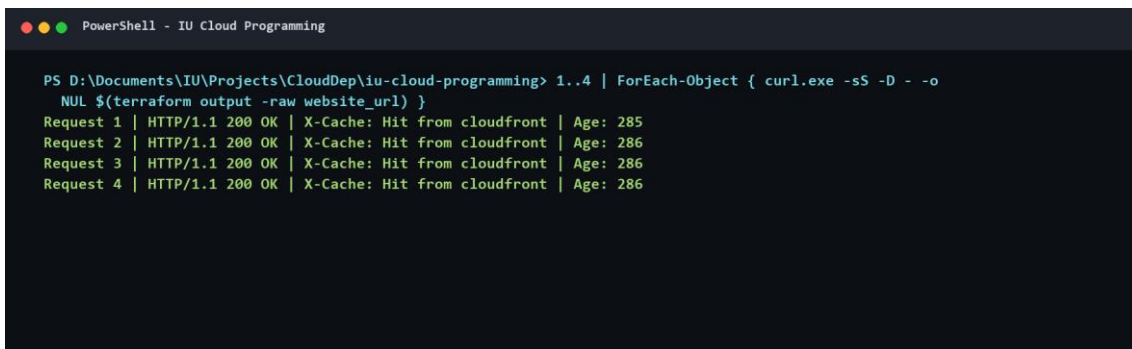
Appendix C. Complete Evidence Gallery (continued)



```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> aws cloudfront get-distribution --id
$(terraform output -raw cloudfront_distribution_id) --query "Distribution.Status"
Deployed
```

Figure 15. CloudFront CLI status is Deployed.



```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> 1..4 | ForEach-Object { curl.exe -sS -D - -o
NUL $(terraform output -raw website_url) }
Request 1 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 285
Request 2 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 286
Request 3 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 286
Request 4 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 286
```

Figure 16. Repeated requests demonstrate Hit from cloudfront.

Appendix C. Complete Evidence Gallery (continued)

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> terraform plan
aws_cloudfront_origin_access_control.site: Refreshing state... [id=E3BY7JC5LZR53Y]
aws_s3_bucket.site: Refreshing state... [id=iu-cloud-programming-spa-943865900195]
aws_s3_bucket_ownership_controls.site: Refreshing state... [id=iu-cloud-programming-spa-943865900195]
aws_s3_bucket_public_access_block.site: Refreshing state... [id=iu-cloud-programming-spa-943865900195]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration
and found no differences, so no changes are needed.
```

Figure 17. Terraform reports no changes after deployment.

```
PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> terraform plan -out=tfplan
Plan: 0 to add, 1 to change, 0 to destroy.

PS ...> terraform apply tfplan
aws_s3_object.index: Modifications complete
Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

PS ...> aws cloudfront create-invalidation --distribution-id EZW53YPRFUFEA --paths "/*"
Invalidation ID: I22L1B6ZRC0RIIX8HM790ZQ5C1
Initial status: InProgress
Final status: Completed

PS ...> curl.exe https://d216eikm4r4k38.cloudfront.net
LIVE CONTENT VERIFIED: Deployment status: Repeatable Terraform update verified.

PS ...> terraform plan
No changes. Your infrastructure matches the configuration.
```

Figure 18. Repeatable object update, invalidation, live verification, and final no-change plan.

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